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Modeling the functioning of rubber trees (*Hevea brasiliensis*) under changing climate and management-driven disturbances

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Rubber plantations are a dominant component of tropical forest landscapes in Southeast Asia and represent a transitional forest system influenced by both climatic variability and management practices. Environmental variability, climate change, and tapping impose growing constraints on rubber plantation productivity and resilience, highlighting the need to access carbon and water dynamics of rubber trees under combined climatic and management-driven disturbances.

We adapted and extended the process-based CASTANEA-MAESPA model to simulate the biophysical functioning of mature rubber plantations in Thailand, with a particular focus on carbon allocation, growth and latex yield under repeated management disturbances associated with latex tapping. A newly developed latex production module was integrated to explicitly link tapping practices to whole-tree carbon balance, growth, and ecosystem fluxes. Model simulations were evaluated against observed water and carbon fluxes and stocks at the Chachoengsao Rubber Research Center, as well as water fluxes and latex yield data from the Sithiporn Kridakorn Research Station. The model reproduced observed fluxes and stocks with good accuracy across sites, capturing the sensitivity of growth and latex yield to environmental conditions. We explored the impact of future climate on modelled latex yield and rubber growth. The insight from that future climate scenario analysis shows the importance of future rising CO_2, which may partially offset the negative effects of climate change on rubber productivity in marginal rubber-growing areas of Southeast Asia, but these gains could be limited by unaccounted future biotic stresses or other processes not included in the model. Overall, this study demonstrates how process-based modelling can be used to provides insights into the resilience and the functioning of rubber plantations within rapidly transitioning tropical landscapes.